Predicting Total Medicare Beneficiaries by State (2023)

Team G11: Sidharth Dusanapudi, Ashish Sunkarapalli, Jayden Sincere Brown, Krish Venkatram Mallina

1. Introduction + Data Description

- This project studies why total Medicare beneficiary counts differ across U.S. states in 2023.
- Medicare is a major part of the U.S. healthcare system, so state differences may reflect aging, population size, and healthcare need (CMS, 2025; U.S. Census Bureau, 2024).
- A key benefit of the analysis is that it compares demographic, health, socioeconomic, access, and political variables in one state-level model.
- The unit of analysis is the state, with one observation for each of the 50 states, and the results describe association rather than causation.

Research Questions

- We hypothesize that states with a higher average age and a Democratic political affiliation will have fewer total Medicare beneficiaries compared to Republican states.
- We hypothesize that states with greater healthcare provider availability (physicians per 100,000 residents) will have higher total Medicare beneficiary counts.
- We hypothesize that states with higher median family incomes will have lower total Medicare beneficiary counts.

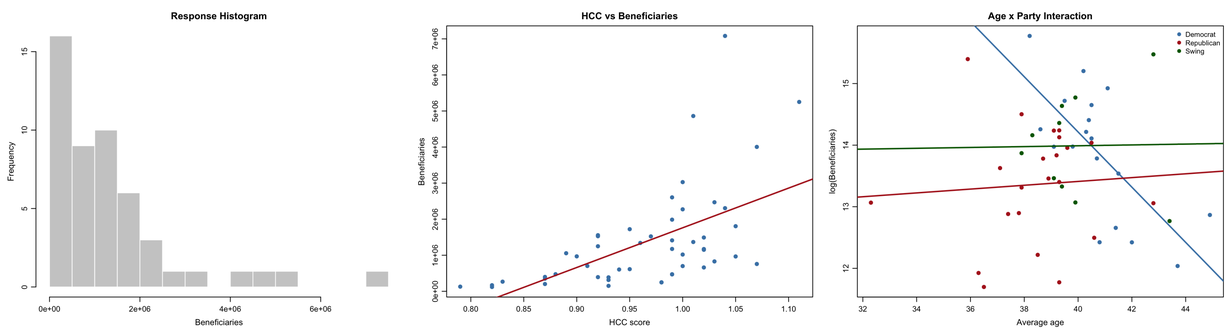
Data Description

- The final data set contains one 2023 observation for each of the 50 states, with Medicare counts from CMS and additional predictors from Census/ACS, Becker's ASC Review, and 270toWin.
- Quantitative predictors: average age, HCC score, percent male, and median family income.
- Qualitative predictors: political affiliation, education quartile, physician quartile, and population quartile.
- Education, physician availability, and population were converted to quartiles; AK, AL, AR, and AZ population values were corrected before quartile creation.
- This design lets us compare broad state-level patterns rather than individual Medicare use.

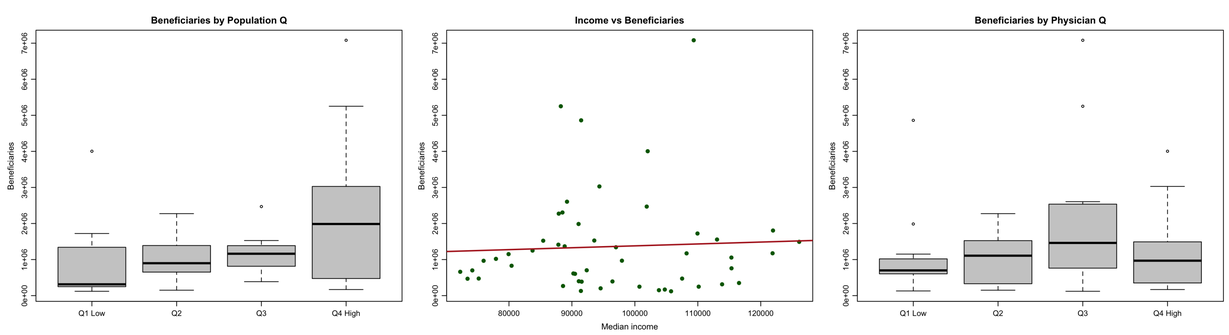
Variable	Role	Levels / Units
Beneficiaries	Response	People
Avg_Age	Quant	Years
HCC_Score	Quant	Risk index
Pct_Male	Quant	Proportion
Median_Income	Quant	U.S. dollars
Political_Affiliation	Qual	Democrat, Republican, Swing
Education_Q	Qual	Q1 Low, Q2, Q3, Q4 High
Physician_Q	Qual	Q1 Low, Q2, Q3, Q4 High
Population_Q	Qual	Q1 Low, Q2, Q3, Q4 High

2. EDA

- The response was strongly right-skewed: mean = 1,360,708, median = 994,359, and range = 120,280 to 7,080,940.
- HCC score had the clearest quantitative relationship with the response ($r = 0.562$), population quartile was the strongest qualitative predictor, and age by political affiliation showed the strongest interaction pattern.
- Median income and physician quartile were visually weaker, so they needed formal testing in the regression model.
- Percent male and education quartile were also reviewed during EDA and kept available for modeling even though they are not plotted here.



Top row: the response histogram, the strongest quantitative predictor, and the clearest interaction pattern.



Bottom row: the strongest qualitative separation plus the weaker income and physician patterns that were carried forward for formal testing.

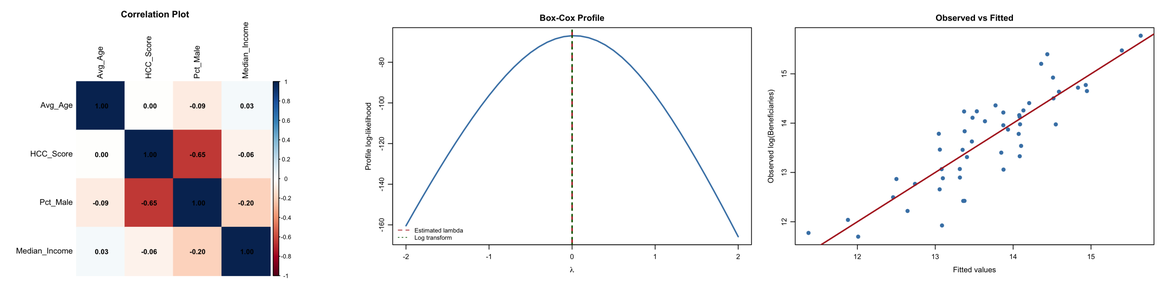
3. Variable Selection + Model Building

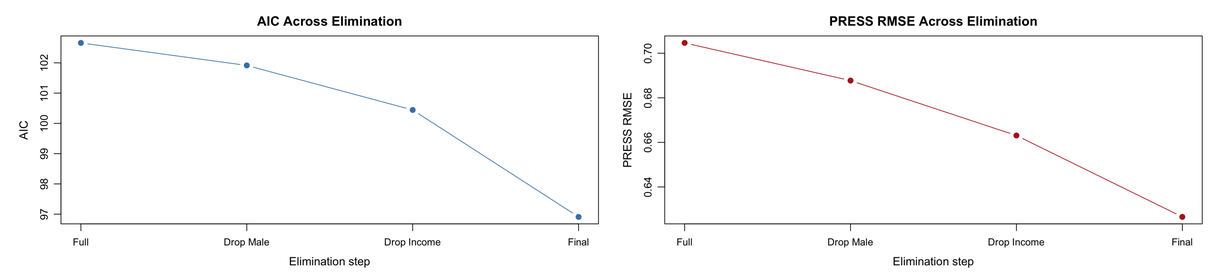
Variable Selection + Multicollinearity

- Screening combined EDA, interaction tests, and multicollinearity checks; the largest VIF in the retained interaction model was 5.63, so collinearity was moderate but acceptable.
- HCC score, population quartile, and age x political affiliation emerged as the most important variables for modeling.

Model Building + Added Technique

- We began with an additive model using all 8 predictors and tested age x political affiliation, HCC x physician quartile, and income x population quartile.
- Nested F tests were used for interaction and grouped-term comparisons, t tests were used for individual coefficients, and the global F test evaluated the final reduced model.
- Only age x political affiliation improved fit ($F = 5.825$, $p = 0.007$), so backward elimination then removed percent male, median income, and physician quartile.
- Box-Cox was the added technique; it supported a log response transformation with $\hat{\lambda} = 0$ and support interval [-0.2, 0.2], which improved the modeling scale.





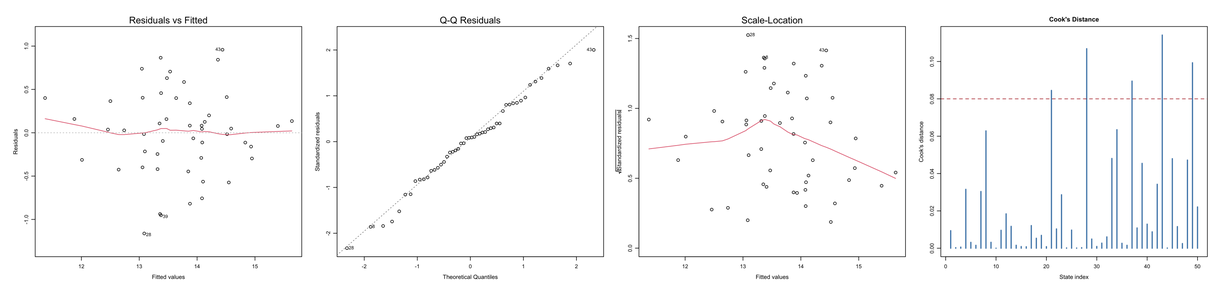
4. Residual Analysis + Final Findings

Residual Analysis

- Linearity, constant variance, and normality looked reasonable after the log transform; residual plots showed no severe pattern and $\mathrm{cor}(|e|, \hat{y}) \approx -0.105$.
- NH, ME, and UT had the highest leverage, while TX and ND had the largest Cook's distances; no single state dominated the fit (max Cook's D ≈ 0.114).

Final Model Assessment

- The final model retained age, political affiliation, age x political affiliation, HCC score, education quartile, and population quartile.
- Model quality was supported by adjusted $R^2 = 0.679$, residual standard error = 0.560, PRESS RMSE = 0.627, and a highly significant global F test ($p < 0.001$).



5. Key Takeaways + Future Work

Conclusion + Interpretation

- The strongest predictors of Medicare beneficiary totals were HCC score, population quartile, and the interaction between age and political affiliation.
- The first research question was supported: the relationship between age and Medicare beneficiary totals differed by political affiliation rather than following one common slope across all states.
- Holding other variables fixed, a 0.10 increase in HCC score corresponded to about 151% higher expected beneficiary totals, and states in population quartile 4 had about 108% higher expected totals than quartile 1 states.
- Physician availability and median income were not supported after adjustment, which means research questions 2 and 3 were not supported in the final model.
- Example prediction: Florida had a predicted total of about 4,856,324 beneficiaries versus an observed total of 5,250,841.

Future Research

- Future work could extend the analysis to multiple years or county-level data and add variables such as percent age 65+, disability prevalence, rurality, hospital capacity, and Medicare Advantage enrollment (CMS, 2025; U.S. Census Bureau, 2024).

6. Sources

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